

Sri Lanka Institute of Information Technology



Lab Sheet 01 2026-Y2-S1-MLB-B4G1-06 Web Based Video Browsing System

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B.Sc. (Hons) in Information Technology

Summary

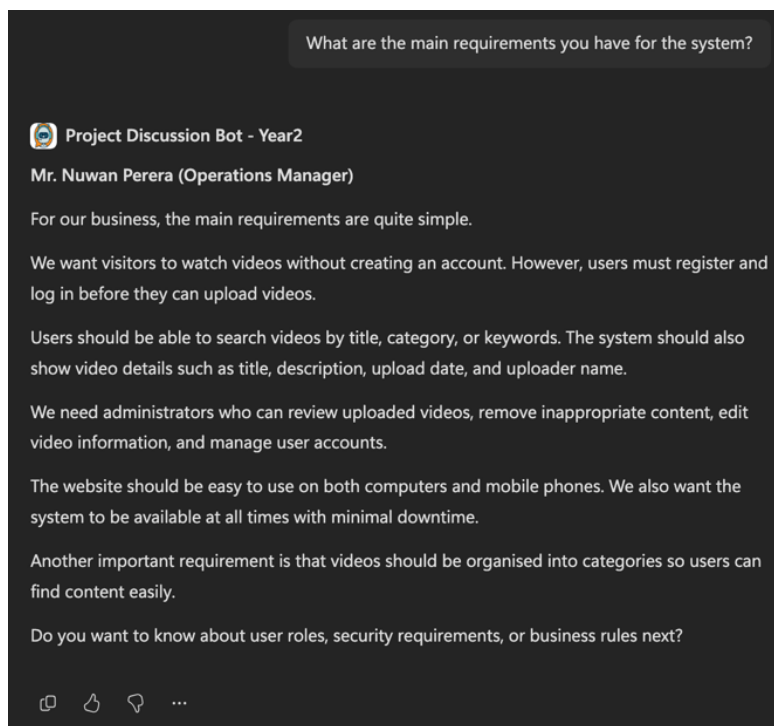
Our team conducted a structured requirements gathering session using the stakeholder simulation bot representing LankaView Media (Pvt) Ltd. Throughout the meeting, we interviewed five distinct client representatives across various departments to ensure a complete understanding of both technical and business needs.

Operations & Business Vision: Conversations with Mr. Nuwan Perera and Mr. Chamath Wickramasinghe established our foundational business rules and user roles. The primary goal is to launch a secure, scalable web platform that permits free viewing while requiring registration for uploads. They stressed keeping the initial release focused on core MVP features to ensure stability and low operational costs.

User Experience & Platform Growth: Ms. Tharushi Fernando and Ms. Dilki Ranathunga provided actionable design expectations. The user interface must be clean, responsive, and easy to navigate on both desktop and mobile devices. Key UX requirements include smooth video playback options, clear upload forms with progress feedback, dedicated support pages, and engagement mechanisms like likes, comments, and homepage features (Trending, Featured, Categories) to keep viewers coming back.

System Administration & Moderation: Mr. Isuru Jayasinghe defined administrative workflows essential for daily maintenance. Administrators require a centralized dashboard featuring overall platform metrics, content moderation tools to act on reported videos, category configuration, and reporting facilities. Upload guardrails (e.g., maximum file sizes and supported formats) were also defined to control storage consumption.

In conclusion, the discussion gave our group clear direction. We synthesized the gathered data into six core system functions; User Management, Video Browsing & Watching, Search & Discovery, Upload & Management, User Engagement, and Administration & Moderation, which directly form the baseline scope for our software implementation phase.



what do you expect for user to experience when visiting the website?



Project Discussion Bot - Year2

Ms. Tharushi Fernando (Customer Experience Officer)

When a user visits our website, I want them to feel that it is simple and easy to use. They should be able to find and watch videos without confusion.

The homepage should clearly show popular videos, new uploads, and video categories.

Searching for videos should be quick and give accurate results.

Video pages should load smoothly and provide a good viewing experience. Users should be able to see the video title, description, uploader name, and upload date easily.

For users who want to upload videos, registration and login should be simple and not take too much time.

I also want the website to work well on mobile phones because many of our users will access it from their phones.

From customer feedback, people usually like:

- Fast page loading
- Easy navigation
- Clear video information
- Simple search
- Fewer steps when uploading videos

Is there anything specific you would like to know about registration, searching, or the video watching experience?

Yes, can we know about user roles, security requirements and business rules



Project Discussion Bot - Year2

Mr. Nuwan Perera (Operations Manager)

Good question. These are important for our business.

User Roles

We need three main user roles.

A Visitor can watch videos, search videos, browse categories, and view video details.

A Registered User can do everything a visitor can do, and also upload videos, edit their own video information, and manage their own profile.

An Administrator can manage users, review uploaded videos, edit video information, remove unsuitable content, and view system reports.

Security Requirements

Users must log in with a username or email and password before uploading videos.

Passwords should be protected and not visible to anyone.

Only administrators should have access to management functions.

The system should prevent unauthorised users from changing or deleting content.

We also want users to be able to report inappropriate videos so administrators can review them.

Business Rules

A user must register before uploading videos.

Each uploaded video must have a title, description, and category.

Users can edit or delete only their own videos.

Administrators can remove any video that violates company policies.

Videos that contain harmful, illegal, or inappropriate content should not be allowed on the platform.

The system should keep a record of upload dates and uploader information for every video.

Is there anything else you would like to know about the business process or daily operations of the system?

tell me about user's profile page and video upload page



Project Discussion Bot - Year2

Ms. Tharushi Fernando (Customer Experience Officer)

That is an important part of the website.

For the **User Profile Page**, I would like users to see their basic information such as name, username, profile photo, and join date. They should also be able to edit their profile details and change their password.

The profile page should show:

- User's uploaded videos
- Total number of uploaded videos
- Total views received
- Favourite or liked videos
- Recent activity

This helps users keep track of their content and activity on the platform.

For the **Video Upload Page**, I want the process to be very simple. Users should be able to select a video file, enter a title, add a description, and choose a category.

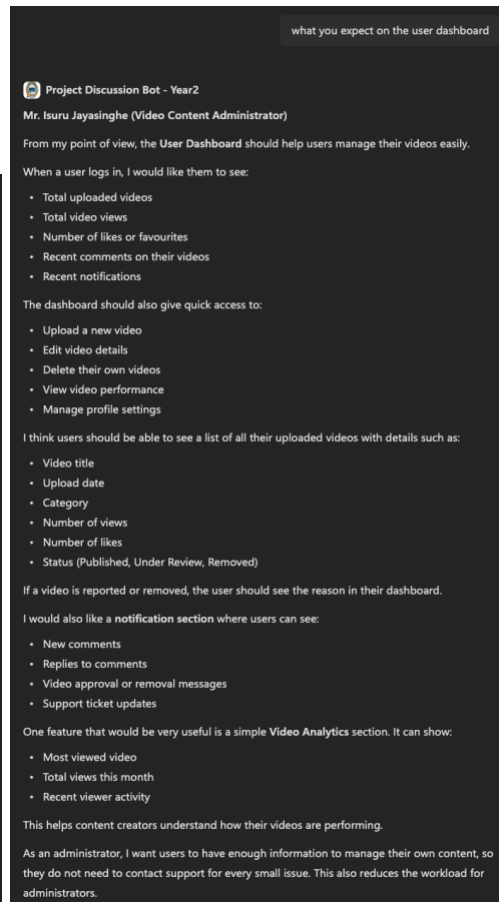
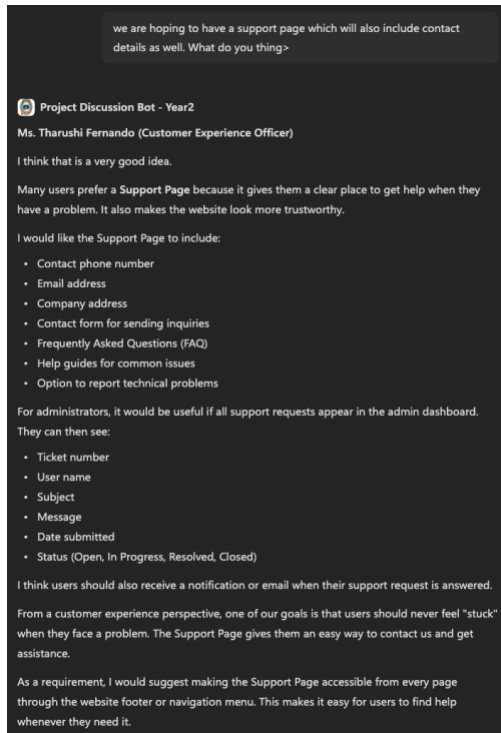
The upload page should also allow users to:

- Add tags or keywords
- Upload a thumbnail image
- See upload progress
- Edit video details after uploading
- Delete their own videos if needed

After a successful upload, the user should receive a confirmation message so they know the video has been submitted successfully.

From a customer experience point of view, the upload process should require as few steps as possible because users usually dislike complicated forms.

Do you want to know about notifications and email messages that users should receive from the system?



Users and their roles

A Visitor: watch videos, search videos, browse categories, and view video details

A Registered User: watch videos, search videos, browse categories, and view video details, upload videos, edit their own video information, and manage their own profile

Administrator: manage users, review uploaded videos, edit video information, remove unsuitable content, and view system reports

Core Functions

1. User Management
2. Video Browsing & Watching
3. Video search & Discovery
4. Video Upload & Management
5. User Engagement
6. Administration & Content Moderation

1.0 User Management

Handles user identity, authentication, access control, and account governance to ensure secure system entry across all user tiers (Visitors, Registered Users, and Administrators).

Account Registration:

- Self-service registration form collecting Full Name, Username, Email Address, and Password.
- Form validation ensuring unique email/username and password strength criteria.
- Automatic confirmation email sent upon successful registration

Authentication & Session Management:

- Secure login via email/username and encrypted password.
- Secure token/session management handling persistent login and explicit logout functionality.
- Role-Based Access Control (RBAC) dynamically restricting endpoints/pages based on user permissions (Visitor vs. Registered Creator vs. Admin).

Profile Management:

- User profile view showcasing basic details (profile photo, bio, join date).
- Editable profile fields (update display name, profile photo, bio) and account settings.
- Overview of user activity: total uploaded videos, total views received, and saved favourite videos.

Password Recovery & Security:

- "Forgot Password" workflow generating time-sensitive password reset links via email.
- Security notification alerts sent via email for password changes or unrecognized login attempts.

2.0 Video Browsing & Watching

Provides seamless video consumption capabilities, allowing both public visitors and authenticated users to stream content effortlessly across devices.

Unregistered Public Access:

- Open access allowing visitors to watch videos, view descriptions, and explore public content without creating an account.

HTML5 Custom Video Player Controls:

- Standard playback controls: Play, Pause, Seek/Progress bar, Rewind, Fast Forward.
- Volume control slider and full-screen toggle.
- Video quality selection (e.g., Auto, 360p, 720p, 1080p) adapting to network condition.

Video Metadata Display:

- Watch page details including Video Title, Description, Upload Date, Category, Tags, View Count, Like Count, and Uploader Profile.

View Tracking & Analytics Trigger:

- Server-side tracking logic to increment the video view count upon playback initiation/completion while preventing artificial view inflation.

3.0 Video search & Discovery

Facilitates fast content retrieval through indexed searching, category hierarchies, and automated discovery feeds to boost platform navigation.

Multi-Parameter Search Engine:

- Global search bar accessible from top navigation on every page.
- Keyword search matching across Video Title, Description, Category, Tags, and Uploader Name.

Category Hierarchy & Tag Navigation:

- Browse videos organized under structured main categories (Education, Tech, Music, Sports, Gaming, etc.).
- Clickable tags/keywords on video pages enabling filtered browsing of similar topics.

Dynamic Content Feeds (Homepage):

- Featured Banner: Highlighting promoted or selected high-quality videos.
- Trending Videos: Algorithmically showcasing videos with high recent views and likes.
- Most Viewed & Recently Uploaded: Automated lists sorted by view counts or chronologically.

Recommendation Engine:

- Contextual "Related Videos" section on the watch page based on matching categories or shared tags.
- Personalized recommendations on the homepage for logged-in users based on viewing preferences.

4.0 Video Upload & Management

Empowers content creators with simple tools to publish video media while enforcing server-side validation and file constraints to control storage usage.

File Upload & Validation Guardrails:

- Multi-format support accepting .MP4, .MOV, and .AVI containers.
- File size validation rejecting uploads exceeding maximum limits (500 MB – 1 GB).
- Duration check restricting video length to a maximum of 30 minutes.
- Daily rate-limiting (e.g., max 10 uploads per user per day).

Metadata Input & Processing:

- Upload form requiring mandatory fields: Title, Description, and Category selection.
- Optional fields: Custom tag input and custom thumbnail image upload.
- Interactive UI upload progress bar with instant confirmation upon completion.

Creator Content Dashboard:

- Management table displaying all owned uploaded videos with status indicators (Published, Under Review, Removed).
- Metrics per video: upload date, views, likes, and comment counts.

Content Editing & Deletion:

- Ability for creators to update video metadata (title, description, thumbnail, category) post-upload.
- Option for creators to permanently delete their own uploaded videos from the platform.

5.0 User Engagement

Drives community interaction, platform feedback, and social sharing to increase overall user retention and content reach.

Interactive Reactions:

- One-click "Like" button with live counter updates.
- "Add to Favorites" action enabling users to save videos to a private collection on their profile.

Comment System:

- Public comment section under each video allowing logged-in users to post comments.
- Notification updates sent to uploader when a new comment is posted.

Social Sharing:

- One-click share buttons generating quick direct URL links to share videos externally.

User Feedback & Issue Reporting:

- "Report Video" mechanism allowing users to flag inappropriate content specifying a violation reason.
- Dedicated Support Page with a contact form, FAQ section, and inquiry submission.
- Support ticket status updates (New, Open, Under Review, Resolved) sent via in-app/email notifications

6.0 Administration & Content Moderation

Gives platform administrators complete administrative control, automated governance, analytics monitoring, and safety moderation.

Centralized Admin Dashboard:

- Executive overview widget showing key platform KPIs: Total Users, Total Uploads, Daily Views, Pending Reports, Open Support Tickets, and Urgent Alerts.

Content Moderation Workflow:

- Queue of user-reported videos displaying reporter reasons and video details.
- Action panel allowing admins to Approve/Keep Video, Send Warning to Uploader, Remove Video, or Suspend User Account.
- Audit logging: System records administrator ID, date/time, and justification for every moderation action.

User & Category Management:

- User directory to view profiles, search accounts, monitor upload histories, or suspend/delete violating accounts.
- Category administration tools: Create new categories, edit names, merge duplicate categories, or correct wrongly categorized videos.

Analytics & Administrative Reporting:

- Exportable reports filterable by date ranges (Today, 7 days, 30 days, Custom).
- Reports include: User growth trends, uploads by category, top viewed/liked content, moderation logs, and support ticket resolution metrics.